

Hybrid MCDM and Simulation-Optimization for Strategic Supplier Selection

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Abstract

Supplier selection for strategic items requires a thorough framework dealing with the qualitative and quantitative aspects of a company's competitive priorities. To address these aspects, this study proposes a model with a multi-sourcing, taking into account multi-criteria, incorporating uncertainty of decision-makers' judgment and supplier-buyer parameters, and integrating with inventory management. We develop a novel two-phase solution approach based on integrated multi-criteria decision-making (MCDM) and multi-objective simulation-optimization (S-O). MCDM methods, including fuzzy AHP and interval fuzzy TOPSIS, are applied to calculate suppliers' scores, incorporating qualitative decision makers' judgment. S-O then combines the (quantitative) cost-related criteria and considers supply disruptions and uncertain supplier-buyer parameters. By running this approach on data generated based on previous studies, we evaluate the impact of decision makers' weight and objectives weight, which are considered to play an important role in supplier selection.

Keywords: Supplier selection, inventory management, multi-criteria decision-making, simulation-optimization, uncertainty, supply disruptions

1. Introduction

Sourcing strategic items can have a significant impact on a company's profit and should therefore be undertaken from the right suppliers, with the right price and quantity, and at the right time. Selecting the right supplier relies on several processes, such as identification of criteria (de'Boer et al., 2001; Aissaoui et al., 2007), which are typically conflicting (Weber et al., 2000). A set of various criteria composed of qualitative and quantitative should be considered when evaluating suppliers (related to the main competitive priorities i.e., price, quality, delivery, flexibility, relationship, and service) (Yadav & Sharma, 2016).

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Furthermore, the complexity of supply and rapid change of the global market have compelled companies to focus on risk mitigation. Mitigating risk is crucial for strategic items since the impact can be tremendous to the entire supply chain's operations. Some of the potential supply risks might come from suppliers due to delivery failures, quality problems, discontinuity of supply, or disruptions (Zsidisin, 2003). To create supply chain resilience, supplier selection processes have to be redesigned. For instance, the adoption of risk-related selection criteria (Igoulalene et al., 2015; Rajesh & Ravi, 2015; Awasthi et al., 2018) and multi-sourcing (Haleh & Hamidi, 2011), as well as the integration of inventory management (Saputro et al., 2020; Firouz et al., 2017; Keskin et al., 2010) can be important levers for risk mitigation in supplier selection.

In practice, evaluating suppliers under multiple criteria is typically performed based on decision-makers' judgment (DMs). This can lead to vague judgment when the exact values of the evaluated alternatives are not available. In this uncertain decision environment, DMs' opinions or judgments need to be perceived realistically to avoid potentially misleading decision-making. It requires transforming linguistic variables into uncertain numerical values (i.e, fuzzy or interval) (Haeri & Rezaei, 2019).

Several studies have attempted to tackle supplier selection problems under multi-criteria evaluation. Most of these studies have applied multi-criteria decision-making (MCDM) approaches, including analytical hierarchy process (AHP), analytic network process (ANP), and Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) (Chai et al., 2013). Although these methods can handle various criteria, unfortunately, a standalone MCDM method cannot properly evaluate the implications of multi-sourcing. Therefore, several studies have employed a two-phase solution approach, starting with MCDM and then optimizing order allocation of multiple suppliers (Singh, 2014; Ayhan & Kilic, 2015; Gören, 2018; Hamdan & Cheaitou, 2017; Kilic & Yalcin, 2020). Still, these studies do not accurately consider risk factors in terms of delivery delay, imperfect quality, and disruptions, and integration of mitigation strategies via inventory management.

Our study has therefore a twofold contribution. First, we propose a comprehensive model by considering both qualitative and quantitative criteria to include risk factors and applying their mitigation strategy (with multi-sourcing and inventory management). The proposed model also addresses the inherent uncertainty to accommodate more realistic DMs' judgment. Second, we develop a novel two-phase solution approach using hybrid MCDM and simulation-optimization to solve the proposed model. Two MCDM methods, namely fuzzy AHP and interval TOPSIS are employed to incorporate DMs' uncertainty in perceiving their opinion when determining the criteria weight and evaluating suppliers, respectively. In addition, the simulation-optimization method can handle uncertain supplier-related parameters (quantitative criteria) and consider disruptions.

The remainder of this paper is organized as follows. Section 2 presents a brief review of relevant literature on supplier selection studies, employing a two-phase solution approach. The problem's context and model formulation are defined in Section 3. Section 4 describes the proposed solution approach, including MCDM and simulation-optimization. In Section 5, an example is given to illustrate the application of the proposed solution approach. In addition, we provide sensitivity analysis on objectives and DMs weights.

2. Literature Review

Studies on supplier selection have grown rapidly in the supply chain management literature. It becomes a critical concern for companies when the selection is focused on purchases that have a strategic role and impact on profitability and operations (strategic items). The problems are diverse in terms of selection criteria, sourcing strategy, decision scope, and decision environment.

Among all these problems, multi-sourcing and multi-criteria are particularly challenging and critical for strategic items in order to mitigate risk of supply and contribute profitability. Recent literature has focused on these problems (e.g., Singh (2014); Ayhan & Kilic (2015); Hamdan & Cheaitou (2017); Cheraghalipour & Farsad (2018); Gören (2018); Kilic & Yalcin (2020)). The respective studies integrated supplier selection with order allocation, subject to uncertainty of DMs' judgment.

The aforementioned studies have proposed a two-phase solution approach dealing with multi-sourcing and incorporating qualitative and quantitative criteria, as well as DMs' judgment uncertainty for a comprehensive decision-making process. Typically, supplier evaluation with respect to the criteria is performed in the first phase of the solution approach to obtain a suppliers' score. In the second phase, final decisions regarding supplier selection and order allocation are determined, considering the suppliers' score.

The two phases are generally approached with MCDM and optimization, respectively. Singh (2014) tackled supplier selection problem using fuzzy TOPSIS and mixed-integer linear programming (MILP). Ayhan & Kilic (2015) proposed an integrated approach using fuzzy AHP to determine criteria weight and MILP to determine supplier selection and order allocation. Hamdan & Cheaitou (2017) applied integrated AHP, fuzzy TOPSIS, and mathematical programming to solve supplier selection and order allocation with a multi-objective model. In the first phase, AHP and fuzzy TOPSIS were used to determine criteria weight and supplier score, respectively. Cheraghalipour & Farsad (2018) focused on integrated supplier selection considering disruption risk. However, they did not consider the impact of disruptions, nor uncertainty was included in the model. The best-worst method was employed to determine the criteria weight and calculate the suppliers' score. The final decisions were determined via revised multi-choice goal programming. Gören (2018) introduced integrated fuzzy decision making trial and evaluation laboratory (DEMATEL), Taguchi loss function, and mathematical programming. Criteria weight and supplier score were respectively calculated using fuzzy DEMATEL and Taguchi loss function. However, suppliers' performance based on a percentage value can be difficult to be perceived and estimated by DMs for intangible criteria using Taguchi loss function. Kilic & Yalcin (2020) integrated intuitionistic fuzzy TOPSIS with fuzzy goal programming to tackle supplier selection in uncertain environment.

Table 1 summarizes the main features of these studies, including decision scope, objectives, and various sources of uncertainty. Some of those studies have addressed the aforementioned aspects, but with some limitations. The solution approach proposed by the studies depicts evaluation redundancy with respect to price or cost addressed in both phases (e.g., Hamdan & Cheaitou (2017); Gören (2018); Cheraghalipour & Farsad (2018)). Our study's

main contribution is to present a comprehensive multi-objective model by incorporating uncertainty of DMs' judgment and supplier-buyer parameters and integrating the decision scope with order allocation and inventory management. We also focus on risk mitigation by considering some risk factors (e.g., disruptions, imperfect quality, and delivery delay) and extensively take advantage of inventory management as a mitigation strategy. Besides, our study also contributes to a novel two-phase solution approach, using MCDM and S-O. More specifically, we proposed fuzzy AHP and interval TOPSIS to deal with qualitative criteria and supplier evaluation under uncertain DMs' judgment. For the final decision-making, S-O is used to optimize the decisions under multi-objectives. Also, it explicitly addresses uncertain supplier-buyer parameters or other quantitative criteria with discrete-event simulation and incorporates the disruptions information to improve the decisions. Therefore, this problem formulation is distinctive from the previous studies, as some criteria that were typically considered qualitative and more abstract are here quantified and simulated.

Simulation is indeed a flexible modeling paradigm, which, combined with optimization (S-O), allows to approach a wide variety of complex systems in uncertain environments (Wang & Shi, 2013), including production planning, transport planning, inventory management, production-distribution planning, and supply chain design (Bang & Kim, 2010). Metaheuristics are most commonly used to address these complex problems, as optimality is frequently unattainable. Combining metaheuristics with simulation can be done in different procedures depending on the simulation purpose and hierarchical structures (Figueira & Almada-Lobo, 2014). Simulation can be used to evaluate the performance of various solutions, refine or extend parameters so that a given analytical model can be enhanced, or generate solutions (Figueira & Almada-Lobo, 2014). Our case is the second, as the analytical model that allows to avoid an excessive number of simulations, and hence save computational time.

Table 1: Problem features of supplier selection

Study	Sourcing Strategy	Integration	Objective	Criteria	Uncertainty	Approach
Singh (2014)	M-S	OA	Single	Quantitative	DMs judgment	MCDM, Optimization
Ayhan & Kilic (2015)	M-S	OA	Single	Qualitative, Quantitative	DMs judgment	MCDM, Optimization
Cheaitou & Hamdan (2017)	M-S	OA	Multi	Qualitative, Quantitative	DMs judgment	MCDM, Optimization
Goren (2018)	M-S	OA	Multi	Qualitative, Quantitative	DMs judgment	MCDM, Optimization
Cheraghalipour et al. (2018)	M-S	OA	Multi	Qualitative, Quantitative	-	MCDM, Optimization
Kilic & Yalcin (2020)	M-S	OA	Multi	Qualitative, Quantitative	DMs judgment	MCDM, Optimization
This study	M-S	OA, I	Multi	Qualitative, Quantitative	DMs judgment & Supplier-buyer parameters	MCDM, Simulation-Optimization

3. Model Development

We study supplier selection integrated with inventory management for a single item and single-period based on a multi-sourcing strategy. We extend the model by incorporating imperfect quality, disruptions, and vehicle capacity. Suppliers are selected by considering multi-criteria classified into two objective functions, namely, maximizing a total value of purchasing (TVP) and minimizing total costs (TC).

We consider a network consisting of m suppliers ($j \in J = (1, \dots, m)$) and one buyer that has n manufacturing plants ($i \in I = (1, \dots, n)$) in different locations. The demand of each plant i , which follows a normal distribution, is met through the material supply from one or more suppliers j that are selected ($X_j = 1$), with certain amounts (Y_{ij}). The full notation of parameters and decision variables is shown in Table 2.

In order to manage inventory, a (Q, R) policy is applied by placing an order with a fixed quantity (Q), as soon as the inventory level drops to or below a reorder point (R). Order quantity (Q_{ij}) and reorder point (R_{ij}) have a specific amount since the order allocation of each plant is specific for each selected supplier (Y_{ij}), the so-called inventory compartmentalization.

Table 2: Input parameters and decision variables

Notation	Description
Indices	
i	: index for plant, $i = 1, 2, \dots, n$
j	: index for supplier, $j = 1, \dots, m$
Parameters	
$E[D_i]$	: Expected annual demand of plant i
a_i	: External failure costs per unit for imperfect items of plant i
o_i	: Setup costs of plant i
h_i	: Holding costs per unit for perfect items of plant i
h'_i	: Holding costs per unit for imperfect items of plant i
s_i	: Shortage costs per unit and per time of plant i
SS_j	: Score of supplier j , which refers to a closeness coefficient CC_k
f_j	: Fixed annual contractual costs of supplier j
c_j	: Purchasing costs per unit of supplier j
k_j	: Rate of imperfect quality for supplier j
b_j	: Annual supply capacity of supplier j
u_j	: Capacity of a TL vehicle for supplier j
θ_j	: Disruption frequency rate for supplier j
v_j	: Disruption length for supplier j
$E[LTD_{ij}]$	: Expected lead time demand between plant i and supplier j
$\eta[LTD_{ij}, R_{ij}]$	: Standardized loss function between plant i and supplier j
p_{ij}	: Fixed transportation costs per replenishment from supplier j to plant i
r_{ij}	: Transportation costs per mile and per replenishment from supplier j to plant i
d_{ij}	: Distance between plant i and supplier j
l_{ij}	: Lead time between plant i and supplier j
Decision variables	
X_j	: 1, if supplier j is selected; 0, otherwise
Y_{ij}	: Purchase amount allocated by plant i to supplier j
Q_{ij}	: Order quantity of plant i to supplier j
R_{ij}	: Reorder point of plant i to supplier j

We also consider supply disruptions and their related risk to the entire supply network, such as delivery delays. When deliveries are delayed due to disruptions, the actual observed lead time and corresponding lead time demand will be higher than the stated lead time. It

is critical to mitigate their impact by avoiding more stock outs through proper inventory management. Thus, we determine the reorder points (R_{ij}) by incorporating an adjusted lead time (l'_{ij}) that takes those disruptions into consideration. This is done through refinements undertaken by the proposed solution approach detailed in Section 4.2.2.

3.1. Total Value of Purchasing

The total value of purchasing (TVP) is the consideration in supplier selection of the maximization of the firm's long-term value. Rather than focusing on pure monetary-based values, TVP focuses on the advantage resulting from every unit purchase allocated to the selected suppliers. Since sourcing experiences from every unit purchase can affect a firm's willingness to buy and perceptions toward suppliers, TVP relies on purchase quantity (Y_{ij}). In this context, TVP is perceived based on the non-monetary criteria, which contribute to an intangible value of advantages. This includes service (C1), relationship (C2), and flexibility (C3).

In order to calculate TVP, we assess suppliers based on the aforementioned criteria. The supplier's performance score (SS_j) is a function of those criteria ($SS_j = CC_k = f(C1_k, C2_k, C3_k)$, where $k = j$) which is derived through multi-criteria decision-making approaches. Finally, TVP is maximized by using the following expression.

$$\text{Max } Z1 \text{ (TVP)} = \sum_i^n \sum_j^m SS_j Y_{ij} \quad (1)$$

3.2. Total Costs

Total costs (TC) are considered monetary-based values, which consist of contractual and purchasing costs (2a), inventory costs (2b-1, 2b-2), transportation costs (2c), external failure, and imperfect holding costs (2d).

$$\text{Min } Z2 \text{ (TC)} = \sum_{j=1}^m f_j X_j + \sum_{i=1}^n \sum_{j=1}^m c_j Y_{ij} \quad (2a)$$

$$+ \sum_{i=1}^n \sum_{j=1}^m \frac{o_i Y_{ij}}{Q_{ij}(1 - E[k_j])} + \sum_{i=1}^n \sum_{j=1}^m h_i \left(\frac{Q_{ij}(1 - E[k_j])}{2} + R_{ij} - E[LT D_{ij}] \right) \quad (2b-1)$$

$$+ \sum_{i=1}^n \sum_{j=1}^m s_i \eta(LT D_{ij}, R_{ij}) \frac{Y_{ij}}{Q_{ij}(1 - E[k_j])} \quad (2b-2)$$

$$+ \sum_{i=1}^n \sum_{j=1}^m \frac{(p_{ij} + r_{ij} d_{ij}) \lceil \frac{Q_{ij}}{u_j} \rceil Y_{ij}}{Q_{ij}(1 - E[k_j])} \quad (2c)$$

$$+ \sum_{i=1}^n \sum_{j=1}^m a_i Y_{ij} E[k_j] + \sum_{i=1}^n \sum_{j=1}^m h'_i Q_{ij} E[k_j] \quad (2d)$$

Fixed contractual costs (f_j) incur once a contract is awarded to the selected supplier. Moreover, each plant has to pay variable purchasing costs (c_j) for the order allocated to a supplier.

Some other costs also have to be paid throughout the supply, including inventory and transportation. More specifically, transportation cost for each delivery is charged according to vehicle capacity (u_j), considering mileage (r_{ij}) and fixed costs (p_{ij}). Total inventory costs are calculated according to setup costs (o_i) and inventory carrying costs (h_i). Additionally, shortage costs incur if stock outs occur at plant ($s_i, i \in I$). Due to their different distance and location, a lead time for each pair of supply (supplier-plant) (l_{ij}) is specific. $\eta(.,.)$ in (2b-2) represents the standard loss function.

The average annual transportation costs (2c) are calculated according to the vehicle capacity u_j . The costs per vehicle are measured based on the fixed vehicle charge (p_{ij}) and mileage costs (r_{ij}). In (2c), d_j represents the distance between suppliers and plants, which is measured according to Euclidean measure associated with suppliers' coordinates d_j and plants' coordinates d_i .

We also consider a quality risk by incorporating suppliers' quality variability and its associated costs. In this regard, the incoming material from a supplier includes a specific rate of imperfect quality (k_j). As a result, plants have to spend a specific holding cost (h'_i) for these imperfect items. Additionally, external failure costs (a_i) incurs due to liability or complaints by customers acquiring imperfect items. The expected imperfect rate ($E[k_j]$) is taken into account as a function of the inventory, transportation, and external failure costs. $E[k_j]$ is computed according to a particular distribution; more specifically, it is perceived as uniformly distributed.

3.3. Constraints

The main constraints regard capacity and demand fulfillment.

Constraint (3) ensures that the order allocated to the selected suppliers Y_{ij} must satisfy the demand in each plant $E[D_i]$.

$$\sum_{j=1}^m Y_{ij} = E[D_i], \quad \forall i \in I \quad (3)$$

Due to the suppliers' capacity constraint, the order allocation Y_{ij} should not exceed their capacity b_j .

$$\sum_{i=1}^n Y_{ij} \leq b_j X_j, \quad \forall j \in J \quad (4)$$

Finally, constraint (5) represents non-negativity and binary decision variables.

$$Y_{ij} \geq 0, \quad Q_{ij} \geq 0, \quad X_j = 0 \text{ or } 1, \quad \forall i \in I, \quad \forall j \in J \quad (5)$$

4. Proposed Approach

In order to solve the problem, a two-phase solution approach is developed integrating MCDM and simulation-optimization. In the first phase, we focus on suppliers' evaluation based on the qualitative criteria. We determine the criteria weight using fuzzy AHP and calculate the supplier score using interval TOPSIS. Then, supplier scores are included into the objective function depicted in Eq. (1), to be optimized at the second phase.

After multi-criteria evaluation, the second phase focuses on solving the multi-objective mathematical model defined in Section 3, integrating supplier selection, order allocation, and inventory management. Simulation-optimization is developed to solve this phase. The two-phase solution approach in this study is illustrated in Figure 1.

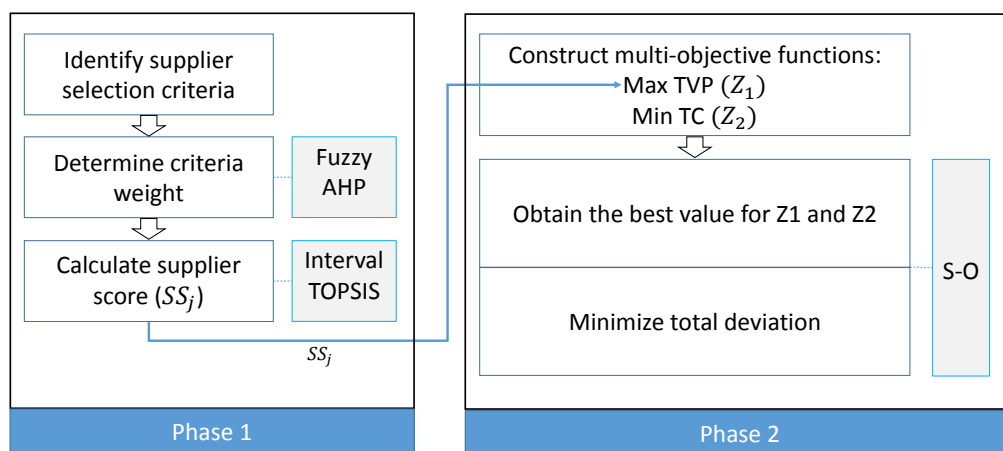


Figure 1: Two-phase solution approach: MCDM and simulation-optimization

4.1. Multi-Criteria Decision-Making

First, criteria weights are determined using fuzzy AHP. Criteria are given different importance by DMs which is perceived using linguistic variables. The linguistic variables are then transformed into its respective triangular fuzzy numbers (TFN) for each Saaty's scale shown in Table 3.

Table 3: Linguistic variables for the importance of the criterion

Linguistic variables	Saaty's Scale	TFN
Equally important	1	1, 1, 1
Weakly or slightly more important	2	1, 2, 3
Moderately more important	3	2, 3, 4
Moderately plus more important	4	3, 4, 5
Strongly more important	5	4, 5, 6
Strongly plus more important	6	5, 6, 7
Strongly very more important	7	6, 7, 8
Very, very strongly more important	8	7, 8, 9
Absolutely more important	9	8, 9, 9

Second, alternatives are evaluated by DMs under each criterion using linguistic variables. Then, DMs judgment is transformed into an interval value shown in Table 4. To calculate the score of alternatives, interval TOPSIS is employed.

Table 4: Linguistic variables for the rating of the alternative

Linguistic Variable	Interval Number
Very Poor (VP)	0, 1
Poor (P)	1, 3
Medium Poor (MP)	3, 4
Fair (F)	4, 5
Medium Good (MG)	5, 6
Good (G)	6, 9
Very Good (VG)	9, 10

4.1.1. Fuzzy AHP

AHP has been widely used for a wide area of decision-making problems due to its advantages: i) it can be used not only to assess relative criteria weights but also to assess the performance of alternatives through pairwise comparisons, ii) it can handle both tangible and intangible attributes, iii) it is suitable for a hierarchical structure of criteria (fundamental components and inter-dependencies) (Zardari et al., 2015).

To deal with qualitative, imprecise information or even incomplete-structures decision problems, fuzzy set theory is employed as a modeling tool for complex systems that can be controlled by humans but are not easy to define exactly. It provides a sensible way to represent vague, ambiguous, and imprecise input of knowledge. Decision makers are usually more confident to perceive interval judgments rather fixed value (crisp) judgments when their opinions can be explicit due to fuzzy nature of evaluation process.

According to fuzzy set theory, crisp values are transformed into fuzzy numbers. A triangular fuzzy number (TFN) is widely used as fuzzy numbers. It involves lower, middle, and upper values.

Definition 1. A fuzzy number M on $R \in (-\infty, +\infty)$ is defined to be a fuzzy triangular number if its membership function $\mu_m : R \rightarrow [0, 1]$ is equal to:

$$\mu_m(x) = \begin{cases} \frac{x}{m-l} - \frac{l}{m-l}, & \text{if } x \in [l, m] \\ \frac{x}{m-u} - \frac{l}{m-l}, & \text{if } x \in [m, u] \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

In Eq. (6) l and u stand for the lower and upper value of fuzzy number M , respectively, and m represents the middle value, where $l \leq m \leq u$. A TFN, expressed in Eq. (6), is denoted as (l, m, u) . The basic operations of TFNs are defined in Table 5.

Table 5: The algebraic operations of fuzzy numbers

Fuzzy Operation	Fuzzy Formula	Calculation Operation
Addition	$\tilde{a}_1 \oplus \tilde{a}_1$	$(l_1 + l_2, m_1 + m_2, u_1 + u_2)$
Subtraction	$\tilde{a}_1 \ominus \tilde{a}_1$	$(l_1 + u_2, m_1 + m_2, u_1 + l_2)$
Multiplication	$\tilde{a}_1 \otimes \tilde{a}_1$	$(l_1.l_2, m_1.m_2, u_1.u_2)$
Division	$\frac{1}{\tilde{a}_1}$	$(\frac{1}{u_1}, \frac{1}{m_1}, \frac{1}{l_1})$

The deficiency of AHP to deal with the imprecision and subjectiveness in the pairwise comparison process has been improved in fuzzy AHP (Demirel et al., 2008). In this study, fuzzy AHP proposed by Chang (1996) is adopted to determine criteria weight.

Let $C = \{C_1, C_2, \dots, C_n\} (j = 1, 2, \dots, n)$ represent the element of supplier selection criteria. Thus, criteria weight is determined according to the following steps:

Step 1. Construct pairwise comparison matrix for each pair of criteria according to the linguistic variables shown in Table 3.

Step 2. Transform the matrix into triangular fuzzy numbers (TFN) (c.f. Table 3) denoted by $M_{gi}^j, j \in N$.

Step 3. Calculate the value of fuzzy synthetic with respect to the i^{th} criterion using

$$S_i = \sum_{j=1}^n M_{gi}^j \otimes \left[\sum_{i=1}^m \sum_{j=1}^n M_{gi}^j \right]^{-1} \quad (7)$$

where

$$\sum_{j=1}^n M_{gi}^j = (\sum_{i=1}^m l_i, \sum_{i=1}^m m_i, \sum_{i=1}^m u_i)$$

$$\left[\sum_{i=1}^m \sum_{j=1}^n M_{gi}^j \right]^{-1} = \left(\frac{1}{\sum_{i=1}^m u_i, \sum_{i=1}^m m_i, \sum_{i=1}^m l_i} \right)$$

Step 4. Determine the degree of possibility of $M_{2(l,m,u)} \geq M_{1(l,m,u)}$ using

$$V(M_2 \geq M_1) = \sup_{y \geq x} [\min(\mu_{M_1}(x), \mu_{M_2}(y))] \quad (8)$$

$$V(M_2 \geq M_1) = hgt(\mu_{M_1} \cap \mu_{M_2}) = \mu_{M_2}(d) = \begin{cases} 1, & \text{if } m_2 \geq m_1 \\ 0, & \text{if } l_1 \geq u_2 \\ \frac{l_1 - u_2}{(m_2 - u_2) - (m_1 - l_1)}, & \text{otherwise} \end{cases}$$

Step 5. Define a convex fuzzy number as

$$V(F \geq F_1, F_2, \dots, F_k) = \min V(F \geq F_i), \quad i = 1, 2, \dots, k \quad (9)$$

$$V(F_i) = \min V(F_i \geq F_k) = W_i', \quad k = 1, 2, \dots, n \text{ and } k \neq i$$

Step 6. Determine the criteria weight vector using

$$W' = (W'_1, W'_2, \dots, W'_n)^T \quad (10)$$

Step 7. After normalization, obtain the priority weights as

$$W = (W_1, W_w, \dots, W_n)^T \quad (11)$$

where W is a crisp number

4.1.2. Interval TOPSIS

TOPSIS is a method based on the concept that the ranking of alternatives is based on the shortest distance from the positive-ideal solution (PIS) and the farthest distance from the negative-ideal solution (NIS) (Hwang & Yoon, 1981). The wide application of TOPSIS in decision-making problems comes from its advantages, including: i) a sound logic that represents the rationale of DM's choice; ii) a scalar value that accounts for both the best and worst alternatives simultaneously; iii) it is not restrained by the human capacity for information processing since DM's evaluation is based on cardinal absolute measurement instead of pairwise comparison; iv) a sensible computation process that can be programmed easily into a spreadsheet (Shih et al., 2007). By using pairwise comparison, consistent judgment becomes very difficult to make when evaluating typically more than seven alternatives since the number of pairwise comparisons increases rapidly with the number of criteria or alternatives ($n(n-1)/2$) (Shih et al., 2007). Therefore, we can use TOPSIS to evaluate a number of suppliers.

Decision-makers would be more comfortable to perceive their opinion into interval measurement when confronting with uncertainty or lack of certain information. According to Jahanshahloo et al. (2009), we adapt interval TOPSIS in this study. Each step of the procedure is explained in the following.

Let $A = \{A_1, A_2, \dots, A_m\} (i = 1, 2, \dots, m)$ be a discrete set of m feasible alternatives, $C = \{C_1, C_2, \dots, C_n\} (j = 1, 2, \dots, n)$ be a finite set of attributes, and $DM = \{DM_1, DM_2, \dots, DM_l\} (k = 1, 2, \dots, l)$ be a group of DMs.

Step 1. For each DM, evaluate each alternative with respect to n attributes using linguistic variables, as shown in Table 4, whose value is an interval ($x_{ij} \in [x_{ij}^{(l)}, x_{ij}^{(u)}]$).

Step 2. For each DM, construct decision matrix which denotes by

$$X_k = ([x_{ij}^{(l)}, x_{ij}^{(u)}])_{m \times n} \quad (12)$$

$$= \begin{matrix} & \begin{matrix} C_1 & C_2 & \dots & C_n \end{matrix} \\ \begin{matrix} A_1 \\ A_2 \\ \vdots \\ A_m \end{matrix} & \begin{pmatrix} [x_{11}^{k(l)}, x_{11}^{k(u)}] & [x_{12}^{k(l)}, x_{12}^{k(u)}] & \dots & [x_{1n}^{k(l)}, x_{1n}^{k(u)}] \\ [x_{21}^{k(l)}, x_{21}^{k(u)}] & [x_{22}^{k(l)}, x_{22}^{k(u)}] & \dots & [x_{2n}^{k(l)}, x_{2n}^{k(u)}] \\ \vdots & \vdots & \vdots & \vdots \\ [x_{m1}^{k(l)}, x_{m1}^{k(u)}] & [x_{m2}^{k(l)}, x_{m2}^{k(u)}] & \dots & [x_{mn}^{k(l)}, x_{mn}^{k(u)}] \end{pmatrix} \end{matrix}$$

Step 3. The weight of k^{th} DMs ($k \in L$) is denoted by vector $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_l)^T$, such that $\lambda_k \geq 0$, $\sum_{k=1}^l \lambda_k = 1$. Given the DMs weight, aggregate the decision matrices into a collective matrix G .

$$G = \sum_{k=1}^l \lambda_k G_k = ([g_{ij}^{(l)}, g_{ij}^{(u)}])_{m \times n} \quad (13)$$

Step 4. Calculate the normalized decision matrix R

$$R_k = ([r_{ij}^{(l)}, r_{ij}^{(u)}])_{m \times n} \quad (14)$$

$$= \begin{matrix} & \begin{matrix} C_1 & C_2 & \dots & C_n \end{matrix} \\ \begin{matrix} A_1 \\ A_2 \\ \vdots \\ A_m \end{matrix} & \begin{pmatrix} [r_{11}^{(l)}, r_{11}^{(u)}] & [r_{12}^{(l)}, r_{12}^{(u)}] & \dots & [r_{1n}^{(l)}, r_{1n}^{(u)}] \\ [r_{21}^{(l)}, r_{21}^{(u)}] & [r_{22}^{(l)}, r_{22}^{(u)}] & \dots & [r_{2n}^{(l)}, r_{2n}^{(u)}] \\ \vdots & \vdots & \vdots & \vdots \\ [r_{m1}^{(l)}, r_{m1}^{(u)}] & [r_{m2}^{(l)}, r_{m2}^{(u)}] & \dots & [r_{mn}^{(l)}, r_{mn}^{(u)}] \end{pmatrix} \end{matrix}$$

We can further transform the aggregated decision matrix $([g_{ij}^{(l)}, g_{ij}^{(u)}])_{m \times n}$ into normalized decision matrix $([r_{ij}^{(l)}, r_{ij}^{(u)}])_{m \times n}$ using the following formula

$$r_{ij}^{(l)} = \frac{g_{ij}^{(l)}}{\sqrt{\sum_{i=1}^m (g_{ij}^{(l)})^2 + (g_{ij}^{(u)})^2}}, \quad \forall i \in M, j \in N \quad (15)$$

$$r_{ij}^{(u)} = \frac{g_{ij}^{(u)}}{\sqrt{\sum_{i=1}^m (g_{ij}^{(l)})^2 + (g_{ij}^{(u)})^2}}, \quad \forall i \in M, j \in N \quad (16)$$

Step 5. Calculate weighted normalized decision matrix R considering the different importance of each attribute as decision matrix V .

$$V = ([v_{ij}^{(l)}, v_{ij}^{(u)}])_{m \times n} = ([w_j r_{ij}^{(l)}, w_j r_{ij}^{(u)}])_{m \times n} \quad (17)$$

$$= \begin{matrix} & \begin{matrix} C_1 & C_2 & \dots & C_n \end{matrix} \\ \begin{matrix} A_1 \\ A_2 \\ \vdots \\ A_m \end{matrix} & \begin{pmatrix} [v_{11}^{(l)}, v_{11}^{(u)}] & [v_{12}^{(l)}, v_{12}^{(u)}] & \dots & [v_{1n}^{(l)}, v_{1n}^{(u)}] \\ [v_{21}^{(l)}, v_{21}^{(u)}] & [v_{22}^{(l)}, v_{22}^{(u)}] & \dots & [v_{2n}^{(l)}, v_{2n}^{(u)}] \\ \vdots & \vdots & \vdots & \vdots \\ [v_{m1}^{(l)}, v_{m1}^{(u)}] & [v_{m2}^{(l)}, v_{m2}^{(u)}] & \dots & [v_{mn}^{(l)}, v_{mn}^{(u)}] \end{pmatrix} \end{matrix}$$

where w_j is the weight of the j^{th} attribute, such that $0 \leq w_j \leq 1$, and $\sum_{j=1}^n w_j = 1$.

Step 6. Find the positive ideal solution (PIS) and negative ideal solution (NIS).

Step 6.1 Determine PIS

Determine the best value of alternative A_k based on the criteria, such as maximum for benefit criteria and minimum for cost criteria. Accordingly, $A_k^{+(u)}$ can be defined as follows:

$$A_k^{+(u)} = \{(v_1^{+(u)}, v_2^{+(u)}, \dots, v_n^{+(u)})\} = \{(\max v_{ij}^{(u)} \mid i \in O), (\min v_{ij}^{(l)} \mid i \in I)\} \quad (18)$$

where O is associated with benefit criteria and I with cost criteria.

Determine the worst value for alternative A_k based on the criteria, such as minimum for benefit criteria and maximum for cost criteria. $A_k^{+(l)}$ can be found using the following form:

$$A_k^{+(l)} = \{(v_1^{+(l)}, v_2^{+(l)}, \dots, v_n^{+(l)})\} = \{(\max_{j \neq i} \{v_{ij}^{(u)}, v_{ij}^{(l)} \mid i \in O\}, (\min_{j \neq i} \{v_{ij}^{(u)}, v_{ij}^{(l)} \mid i \in I\})\} \quad (19)$$

Step 6.2 Determine NIS

$$A_k^{-(u)} = \{(v_1^{-(u)}, v_2^{-(u)}, \dots, v_n^{-(u)})\} = \{(\min_{j \neq i} \{v_{ij}^{(u)}, v_{ij}^{(l)} \mid i \in O\}, (\max_{j \neq i} \{v_{ij}^{(u)}, v_{ij}^{(l)} \mid i \in I\})\} \quad (20)$$

$$A_k^{-(l)} = \{(v_1^{-(l)}, v_2^{-(l)}, \dots, v_n^{-(l)})\} = \{(\min v_{ij}^{(l)} \mid i \in O), (\max v_{ij}^{(u)} \mid i \in I)\} \quad (21)$$

Step 7. Calculate the distance of each individual decision A_k from the PIS ($d_k^{+(l)}$, $d_k^{+(u)}$) and NIS ($d_k^{-(l)}$, $d_k^{-(u)}$) using the n -dimensional Euclidean distance.

$$\begin{aligned} d_k^{+(u)} &= \sqrt{\sum_{i \in I} (v_i^{+(u)} - d_{ik}^{(u)})^2 + \sum_{i \in O} (v_i^{+(u)} - d_{ik}^{(l)})^2} \\ d_k^{+(l)} &= \sqrt{\sum_{i \in I} (v_i^{+(l)} - d_{ik}^{(l)})^2 + \sum_{i \in O} (v_i^{+(l)} - d_{ik}^{(u)})^2} \\ d_k^{-(u)} &= \sqrt{\sum_{i \in I} (v_i^{-(l)} - d_{ik}^{(l)})^2 + \sum_{i \in O} (v_i^{-(l)} - d_{ik}^{(u)})^2} \\ d_k^{-(l)} &= \sqrt{\sum_{i \in I} (v_i^{-(u)} - d_{ik}^{(l)})^2 + \sum_{i \in O} (v_i^{-(u)} - d_{ik}^{(u)})^2} \end{aligned} \quad (22)$$

Step 8. Calculate closeness coefficients ($CC_k^{(l)}$, $CC_k^{(u)}$):

$$\begin{aligned} CC_k^{(l)} &= \frac{d_k^{-(l)}}{d_k^{-(u)} + d_k^{+(u)}} \\ CC_k^{(u)} &= \frac{d_k^{-(u)}}{d_k^{-(u)} + d_k^{+(u)}} \end{aligned} \quad (23)$$

Step 9. Rank the best alternative. We adopt Sengupta's approach in the following.

Calculate the mid-point $m(CC_k)$ and half width of the interval closeness coefficient $w(CC_k)$ using

$$m(CC_k) = \frac{1}{2}(CC_k^{(l)} + CC_k^{(u)}) \quad (24)$$

$$w(CC_k) = \frac{1}{2}(CC_k^{(u)} - CC_k^{(l)}) \quad (25)$$

According to the acceptability function, compare two alternatives a and b as follows:

$$\mathcal{A}_{(<)} = \frac{m(b) - m(a)}{w(b) + w(a)} \quad (26)$$

$\mathcal{A}_{(<)}$ can be interpreted as the first interval to be inferior to the second interval. The term "inferior to" ("superior to") can be defined as "less than" ("greater than"). Decision-makers can select an alternative between two according to the value of $\mathcal{A}_{(<)}$. According to this procedure, the best choice of alternative can stand for the one with a smaller uncertain interval (the half width) if two interval numbers have the same mid-point.

4.2. Simulation-Optimization

4.2.1. Multi-Objective Approach

First, we divide the multi-objective model defined in Section 3 into two single-objective sub-problems. The first sub-problem is defined according to the objective function in Eqs. (1) and constraints in Eqs. (3), (4), and (5). The second sub-problem comprises objective function in Eqs. (1a) - (1d), subject to the same constraints. We solve these two sub-problems separately using an S-O approach to obtain their best solutions, $Z1_{max}$ and $Z2_{min}$, respectively.

Second, the distance method is used to calculate the deviation of the objective function (e) representing the distance from the ideal solution (Z^* , where $Z^* = Z_{max}$ for maximization, $Z^* = Z_{min}$ for minimization).

$$f1 = \frac{Z1_{max} - Z1}{Z1_{max}} \quad (27)$$

$$f2 = \frac{Z2 - Z2_{min}}{Z2_{min}} \quad (28)$$

Finally, we transform both objective functions into a single objective for minimizing total deviation (e) using the weighted comprehensive criterion method (WCCM) (Abdallah et al., 2021). The importance weights (α_1, α_2) is also assigned to each objective function. A single objective function is expressed as follows.

$$\text{Min } e = \alpha_1 \cdot f1 + \alpha_2 \cdot f2 \quad (29)$$

s.t.

$$\text{Eqs. (3), (4), (5)} \quad (30)$$

where $\alpha_1 + \alpha_2 = 1$.

4.2.2. Simulation Enhancing The Analytical Model

We develop simulation-optimization using a genetic algorithm (GA) to search within the solution space. A simulation model provides a thorough evaluation for stochastic input parameters, considering stochastic demand, uncertain imperfect rate, and disruptions. In our proposed approach, the GA optimizes decision variables of X (supplier selection) based on the performance measure (e) formulated in Eq. (29).

Given the value of X , which is randomly selected, we determine order allocation (Y) according to the transportation cost given by (2c) in the objective function, constraints (3) and (4). More specifically, we derived the solution using a transportation method. Then, the optimal order quantity from a plant to a supplier (Q_{ij}) is obtained according to inventory costs given by (2b), (2c), and imperfect items' holding costs (2d). To incorporate vehicle capacity, order quantity is determined by using a heuristic (see Saputro et al. (2020)).

The solutions containing supplier selection (X) and inventory decisions (Q, R) are then passed to the simulation incorporating demand uncertainty and disruptions for objective function evaluation (e). The optimization process is powered by simulation's feedback, which is used to refine the lead-time (L). It is used to address possible delays that result from disruptions. The analytical expression (R, Q) is then enhanced while the lead-time is refined. This simulation-optimization approach is known as an analytic model enhancement (AME). The refinement procedure begins such that, for every randomly selected X , and each replication, the lead time derived from simulation based on the mean value is sent for optimization. According to the refined lead time, the reorder point is recalculated. The performance measure (e) is returned to the optimization according to the decision variables sent to the simulation. The GA then uses this performance measure to optimize the solutions. The illustration of AME is shown in Figure 2.

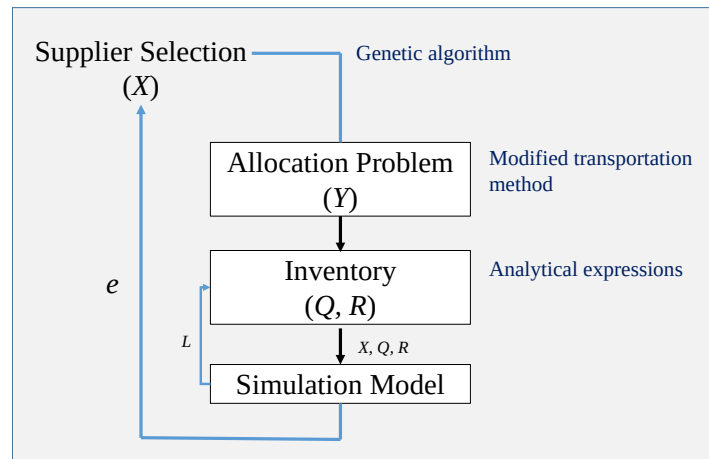


Figure 2: Simulation-optimization: Analytic model enhancement

5. Computational Experiments

In this study, we consider a firm that operates eight manufacturing plants located in different regions. The material supply of each manufacturing plant is sourced from two or

more suppliers. There are ten candidate suppliers to be evaluated under qualitative and quantitative criteria. We use qualitative criteria and their evaluation based on the decision-makers' judgment from Yadav & Sharma (2016)'s study. Quantitative criteria and other parameters indicated in Eqs. (2a)-(2d), (3), and (4) are adopted from Saputro et al. (2020). These quantitative criteria will be assessed objectively in a monetary based-value. The respective quantitative and qualitative criteria are summarized in Table 6. In addition, the values of input parameters, representing fast-moving items, are indicated in Table 7.

Table 6: Quantitative and qualitative supplier selection criteria

Category	Criteria	Sub-criteria	Category	Criteria	Sub-criteria
Quantitative	Cost	Purchasing cost (c)	Qualitative	Service	Technical support
		Contractual cost (f)			Information sharing
		Transportation cost (p, r)			Warranty & claim policy
	Quality	Rate of perfect quality ($1 - k$)		Relationship	Capabilities
		Lead time (l)			Honesty
	Delivery	On-time delivery (<i>simulation</i>)			Reputation
		Vehicle capacity (u)			Trust & partnership
		Distance (d)			Ease of communication
	Technology	Supply capacity (b)		Flexibility	Product mix flexibility
		Disruptions (θ, v)			Volume flexibility
	Risk	Disruptive lead time (<i>simulation</i>)			Process flexibility
		Rate of imperfect quality (k)			Service flexibility

Table 7: Input parameters for fast movers

Parameters	Values	Units
Plant, $i \in I$		
Demand	μ_i, σ_i : U(1000, 3000), U(150, 300)	unit/year
Setup costs	o_i : U(500, 1000)	\$/order
Holding costs	h_i : U(0.5, 3.0)	\$/unit/year
Shortage costs	s_i : U(5.0, 10.0)	\$/unit/year
Imperfect items' holding costs	h'_i : U(0.25, 1.5)	\$/unit/year
External failure costs	a_i : U(0.2, 1)	\$/unit
Location	: [U(0, 500), U(0, 500)]	
Supplier, $j \in J$		
Supply capacity	b_j : U(7500, 10000)	unit
Imperfect rate	k_j : U(0.15, 0.35)	
Vehicle capacity	u_j : U(150, 300)	unit/vehicle
Disruption frequency	θ_j : U(1, 7)	days
Disruption length	v_j : U(0.5, 2)	days
Contractual costs	f_j : U(50000, 100000)	\$
Unit purchasing costs	c_j : U(0.4, 2.0)	\$/unit
Location	: [U(0, 500), U(0, 500)]	
Plant-Supplier, $i \in I, j \in J$		
Fixed transportation costs	p_{ij} : U(250, 500)	\$/order/vehicle
Variable transportation costs	r_{ij} : U(0.75, 3)	\$/mile/vehicle
Lead time	l_{ij} : $\left(\frac{U(1,2)}{60}\right)d_{ij}$	hours

5.1. Suppliers Assessment based on Qualitative Criteria

5.1.1. Determining Criteria Weight

There are 3 criteria and 12 sub-criteria associated with qualitative measures. A decision-maker assessed the criteria and sub-criteria using fuzzy pairwise comparison matrices shown in Table 8 and Table 9, respectively. Finally, the weights are calculated using Fuzzy AHP and the result is shown in Table 5.

Table 8: Fuzzy pairwise comparison matrix among qualitative criteria

Criteria	Service (C1)	Relationship (C2)	Flexibility (C3)
Service (C1)	1, 1, 1	2, 3, 4	1, 2, 3
Relationship (C2)	1/4, 1/3, 1/2	1, 1, 1	1/3, 1/2, 1
Flexibility (C3)	1/3, 1/2, 1	1, 2, 3	1, 1, 1

Table 9: Fuzzy pairwise comparison matrix among qualitative sub-criteria

Service (C1)	Technical support (SC1)	Information sharing (SC2)	Warranty and claim policy (SC3)	Capabilities (SC4)
Technical support (SC1)	1, 1, 1	2, 3, 4	1/4, 1/3, 1/2	1, 2, 3
Information sharing (SC2)	1/4, 1/3, 1/2	1, 1, 1	1/5, 1/4, 1/3	1/4, 1/3, 1/2
Warranty and claim policy (SC3)	2, 3, 4	1, 1, 1	1, 1, 1	2, 3, 4
Capabilities (SC4)	1/3, 1/2, 1	2, 3, 4	1/4, 1/3, 1/2	1, 1, 1
Relationship (C2)	Honesty (SC5)	Reputation (SC6)	Trust & partnership (SC7)	Ease of communication (SC8)
Honesty (SC5)	1, 1, 1	2, 3, 4	4, 5, 6	4, 5, 6
Reputation (SC6)	1/4, 1/3, 1/2	1, 1, 1	2, 3, 4	2, 3, 4
Trust & partnership (SC7)	1/6, 1/5, 1/4	1/4, 1/3, 1/2	1, 1, 1	1, 2, 3
Ease of communication (SC8)	1/6, 1/5, 1/4	1/4, 1/3, 1/2	1/3, 1/2, 1	1, 1, 1
Flexibility (C3)	Product mix flexibility (SC9)	Volume flexibility (SC10)	Process flexibility (SC11)	Service flexibility (SC12)
Product mix flexibility (SC9)	1, 1, 1	1/4, 1/3, 1/2	1, 2, 3	1/4, 1/3, 1/2
Volume flexibility (SC10)	2, 3, 4	1, 1, 1	3, 4, 5	1/3, 1/2, 1
Process flexibility (SC11)	1/3, 1/2, 1	1/5, 1/4, 1/3	1, 1, 1	1/4, 1/3, 1/2
Service flexibility (SC12)	2, 3, 4	1, 2, 3	2, 3, 4	1, 1, 1

Criteria with a high importance weight become a critical aspect of evaluation. According to Table D.21, the DM considers warranty and claim policy as the most critical one for supplier evaluation, followed by technical support, service flexibility, and volume flexibility, respectively.

5.1.2. Determining Suppliers Score

In this stage, suppliers performance are evaluated under sub-criteria using linguistic variables expressed in Table 2. The decision maker judgment regarding suppliers performance is summarized in Table 10. According to this information, the DM' judgments are transformed into their respective interval numbers shown in Table C.14. Supplier score is then determined using interval TOPSIS, and sub-criteria global weights (w_j), indicated in Table 9, are used for this calculation (see Eq. (17) in Section 4.1.2). Finally, supplier score (SS_j) is derived based on the mid-point of the closeness coefficient ($SS_j = CC_k$, where $k = j$), shown in Table 11.

Table 10: Supplier performance under DM's judgment

Supplier	Sub-criteria											
	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12
1	F	MG	F	MG	MP	F	G	P	P	F	MP	MG
2	F	MP	G	F	MP	F	G	F	MP	MP	MP	MP
3	MP	MG	F	MP	MP	F	MP	F	MG	P	P	F
4	F	P	G	F	P	MG	F	MP	F	MG	P	F
5	MP	F	G	F	P	MG	F	F	P	MG	P	P
6	MP	MP	G	MG	MP	F	MP	P	P	F	P	P
7	MP	F	G	F	P	P	F	F	F	P	P	F
8	F	MP	G	G	MP	MG	MG	P	P	MP	P	MP
9	MP	P	MP	MG	P	P	MG	F	F	MP	MP	MG
10	F	MG	MP	G	P	MP	G	F	MP	P	MP	MG

According to the closeness coefficient, supplier 4 has the best qualitative evaluation since it has the highest score. This implies that its overall performance is far from the worst existing evaluation. The second and third best alternatives refer to suppliers 8 and 2, respectively. Supplier 10 represents the worst-performing alternative although its performance on six out of ten criteria is better than supplier 8. This happened mainly due to the criteria weight assigned by the DM. In this study, sub-criteria, including technical support (SC1), warranty & claim policy (SC3), volume flexibility (SC10), and service flexibility (SC12), are given a high priority. At least under one of these sub-criteria (i.e., technical support, warranty & claim policy, and volume flexibility), the performance of supplier 10 underperforms supplier 8.

Table 11: Suppliers score based on the closeness coefficient

Supplier	Closeness Coefficient (CC_k)		
	Interval	Mid-point	Half-width
1	[0.377, 0.685]	0.531	0.154
2	[0.324, 0.814]	0.569	0.245
3	[0.299, 0.532]	0.416	0.117
4	[0.420, 0.842]	0.631	0.211
5	[0.367, 0.760]	0.564	0.197
6	[0.339, 0.769]	0.554	0.215
7	[0.335, 0.757]	0.546	0.211
8	[0.330, 0.836]	0.583	0.253
9	[0.330, 0.539]	0.435	0.104
10	[0.319, 0.544]	0.397	0.112

5.2. Final Selection

The final decision-making for the integrated supplier selection is accomplished by solving the multi-objective model (described in Section 3) using an S-O approach considering disruptions (c.f. Section 4.2). We construct objective function $Z1$, incorporating supplier scores (SS_j) obtained from interval TOPSIS, and objective function $Z2$.

The best values are 14893 and 64972, respectively for $Z1_{max}$ and $Z2_{min}$. For $Z1_{max}$, supplier 4 and 8 are selected. While for $Z2_{min}$, selected suppliers include 4 and 6.

After deriving the best value of each objective, the final solution is derived by minimizing total deviation of both objectives using Eq. (29) and setting up $\alpha_1 = 0.5$ and $\alpha_2 = 0.5$. The best solution was found with a deviation of 0.073 (see Figure B.5). Selected suppliers include 4 and 6. This indicates that $Z1_{max}$ is compromised to achieve the trade-off.

5.3. Sensitivity Analysis

This analysis aims to investigate the impact of DM weight and objective weight. In order to arrive at more general conclusions, we created an additional scenario by using different values of input parameters, as seen in Table E.22. The values represent slow-moving items adopted from Saputro et al. (2020). Additionally, DMs' judgments for suppliers are provided in Appendix C. The analysis is respectively presented in Section 5.3.1 and 5.3.2, emphasizing some key aspects.

5.3.1. Impact of Decision Maker Weight

We investigate the impact of DMs weight (λ_k) on the suppliers score (SS_j) and their associated ranking. Thus, we consider an additional DM, namely DM_1 and DM_2 . To perform this analysis, we just focus on the first phase of the solution approach for fast and slow movers. For each pair of λ_1 and λ_2 , we set up the weight varying from 0.4 up to 0.6, and $\lambda_1 + \lambda_2 = 1$. The results of the sensitivity analysis are shown in Figure 3.

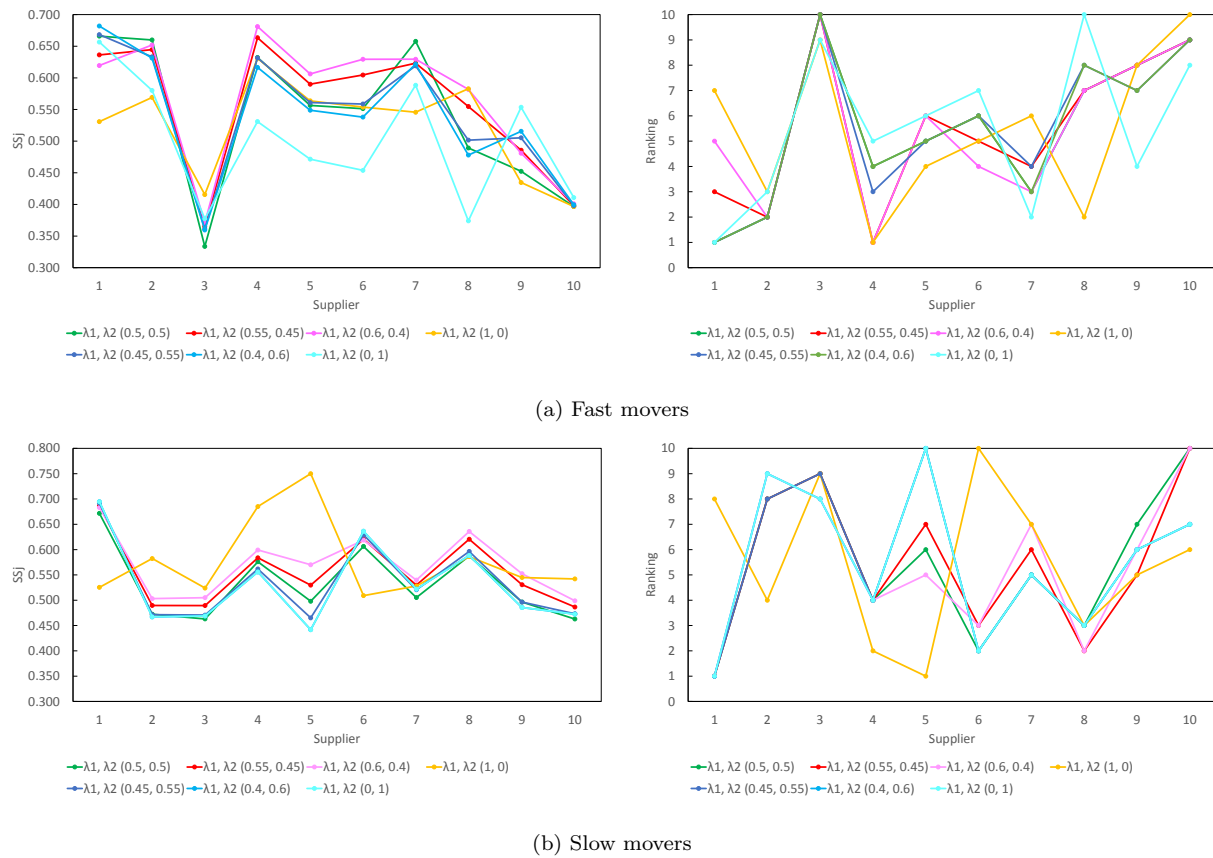


Figure 3: Impact of DMs weight (λ_k) on the suppliers score (SS_j) and ranking

Figure 3 indicates that the opinions of each DM toward suppliers performance can differ. Variation of both suppliers score and ranking is considerable for each DM ($\lambda_1 = 1$ or $\lambda_2 = 1$). For fast movers, DM1 and DM2 make contrast assessment for suppliers 1, 7, 8, and 9. While for slow movers, assessment of suppliers 1, 2, 5, 6, and 9 contrast between DM1 and DM2. Therefore, to accommodate the different opinions, weight needs to be assigned to each DM so that consensus can be achieved.

The result of the sensitivity shows that the impact of λ_k on supplier score and ranking is quite evident. It improves satisfaction degree of consensus for each DM. According to each scenario of DMs weight, small variation can be achieved when $\lambda_1 = 0.45$ and $\lambda_2 = 0.55$ for fast movers, and $\lambda_1 = 0.4$ and $\lambda_2 = 0.6$ for slow movers. Tabel 12 shows the mean variation of suppliers' score according to different scenarios of λ_1 and λ_2 against a single $\lambda(\lambda_1 = 1$ or $\lambda_2 = 1)$.

Table 12: Variation of supplier score according to different scenarios of (λ_k)

Problem	Scenarios				
	$\lambda_1, \lambda_2 (0.6, 0.4)$	$\lambda_1, \lambda_2 (0.55, 0.45)$	$\lambda_1, \lambda_2 (0.5, 0.5)$	$\lambda_1, \lambda_2 (0.45, 0.55)$	$\lambda_1, \lambda_2 (0.4, 0.6)$
Fast movers	0.2920	0.2658	0.2534	0.2230	0.2234
Slow movers	0.2128	0.1953	0.1894	0.1784	0.1728

5.3.2. Impact of Objective Weight

Analysis regarding the impact of objective weight (α_k) is further performed for each problem. The objective weight varies between 0.2 and 0.8 for each pair of α_1 and α_2 . The results of this sensitivity analysis are shown in Figure 4.

Both scenarios indicate that the trade-off between objective functions $Z1$ and $Z2$ decreases when weights are more unbalanced. In those cases, the total deviation of the objectives tends to decrease. By contrast, total deviation increases when the weight of both objectives is nearly the same. Thus, it is not easy to achieve a high yield without compromising another objective.

Furthermore, our analysis indicates that the best trade-off is achieved when $Z2$ is given a priority, resulting in lower deviation compared to other scenarios. The total deviation on average decreases by 22% and 16%, respectively for fast and slow movers, for $\alpha_2 > 0.5$. In other words, minimizing total costs can be considered more important than maximizing the total value of purchasing when selecting suppliers. This is aligned with the insight pointed by the case study explored by Gören (2018).

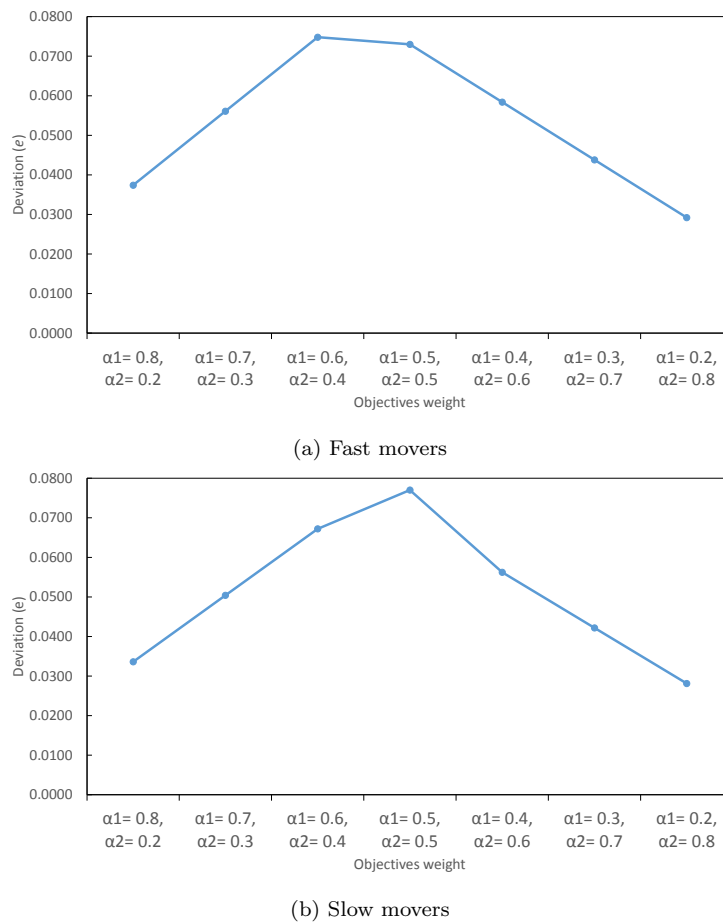


Figure 4: Impact of objective weight (α_k) on the deviation (e)

6. Conclusion

Due to high-profit impact and supply complexity, this study addresses a supplier selection problem incorporating criteria holistically under uncertainty and disruptions risk mitigation. A novel two-phase solution approach is proposed to solve the model with multi-objective. Fuzzy AHP and interval TOPSIS are respectively used to perceive imprecise DMs' judgment in determining weight and assessing suppliers under qualitative criteria. The final decision-making process is performed using S-O based on analytic model enhancement (AME). AME provides a better decision for supplier selection and inventory management since the lead time is refined according to the disruption information. In other words, this solution approach is useful to deal with disruption risk mitigation.

Our analysis draws important implications for decision-makers in supplier selection. Selection criteria should be well incorporated for both qualitative and quantitative. Evaluating suppliers under quantitative criteria should be objectively performed as it can be measured based on a monetary-based value. This monetary measure should be the focus for purchases comprising high-profit impact. The quantitative criteria become a critical aspect since it refers to a firm's core performance (e.g., quality, delivery, and cost). Also, it is associated with risk factors (i.e., disruptions, imperfect quality, delivery delay), which becomes a critical issue for the purchases whose supply complexity is high such as strategic items. It becomes relevant since it also affects inventory decisions (Saputro et al., 2020).

Under uncertainty in which information regarding suppliers can be incomplete or non-obtainable, imprecise or vague judgment raises particularly for qualitative criteria. This can result in contradictory judgment among DMs. Therefore, a weight needs to be assigned to each DM to accommodate the degree of satisfaction. In practice, it is important to look at DMs' knowledge, experience, and consistency when assigning their weights.

A pre-qualification or screening process might be established in supplier selection, particularly when the number of candidates restrains human's evaluation capacity. The two-phase solution approach proposed in this study discloses a comprehensive decision-making process, which does not need pre-qualification. This also enhances the final decision-making by optimizing the decisions jointly via S-O, considering multi-objectives.

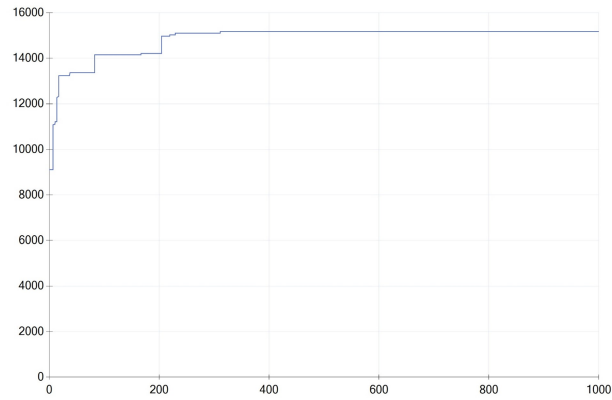
This study has limitations that might lead to interesting future research. Despite the fact that the study has addressed the risk factors (i.g., disruptions, imperfect quality, delivery delay), other risk factors in global sourcing might also exist due to political or social instability. The future study can be extended, considering this aspect to sustain resilient supplier selection in global sourcing. Furthermore, the increase of awareness on sustainability might compel firms to identify related criteria and incorporate them into supplier selection. Developing a framework for supplier evaluation under sustainability could also be an interesting future research direction.

Appendix A. Qualitative Criteria: Strategic Items Suppliers

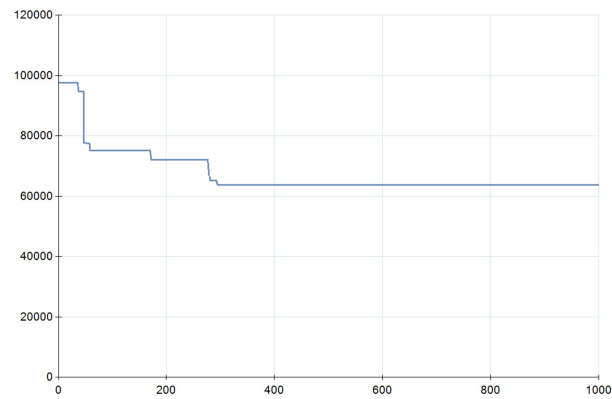
Table A.13: Qualitative criteria for supplier selection

Criteria	Description	Sub-criteria	Description
Service	The after-sales service which promotes customers satisfaction and influences customer purchasing intentions	Technical support	Commitment of a supplier to provide technical support services
		Information sharing	The willingness of a supplier to share technical information
		Warranty and claim policy	The intention of a supplier to provide warranties or agreements between the customer and the supplier for the faulty products
		Capabilities	The capability of a supplier to resolve issues or conflict
Relationship	The buyer-supplier relationship that enhances mutual motivation and results in better development of the total economy	Honesty	The attitude and responsibility of managers in professional relationship
		Reputation	The track record of Supplier indicating a cooperation experience with large enterprises
		Trust & partnership	The commitment s of a supplier to Establish mutually beneficial long-term supplier relationship
		Ease of communication	The ability of supplier in providing an effective commutations system to customers
Flexibility	The ability of a supplier to adapt to external changes while maintaining satisfactory system performance	Product mix flexibility	The ability to change the variety of products produced (customers' orders)
		Volume flexibility	The ability to respond to change in demand
		Process flexibility	The ability to adapt the production technology and its process in order to respond to the new customer product characteristics
		Service flexibility	The ability to handle the abnormal orders without compromising the existing product price

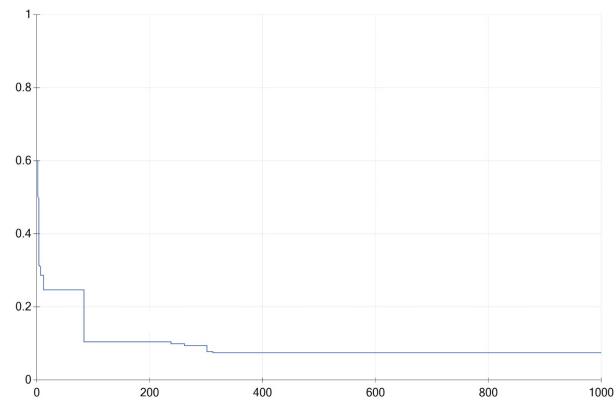
Appendix B. Convergence of Genetic Algorithm



(a)



(b)



(c)

Figure B.5: The convergence under multi-objective settings: (a) Value of purchasing ($Z1$), (b) Total costs ($Z2$), (c) Total deviation (e)

Appendix C. Decision Makers Judgment

Table C.14: Interval values for supplier assessment

Supplier	Sub-criteria											
	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12
1	[4, 5]	[5, 6]	[4, 5]	[5, 6]	[3, 4]	[4, 5]	[6, 9]	[1, 3]	[1, 3]	[4, 5]	[3, 4]	[5, 6]
2	[4, 5]	[3, 4]	[6, 9]	[4, 5]	[3, 4]	[4, 5]	[6, 9]	[4, 5]	[3, 4]	[3, 4]	[3, 4]	[3, 4]
3	[3, 4]	[5, 6]	[4, 5]	[3, 4]	[3, 4]	[4, 5]	[3, 4]	[4, 5]	[5, 6]	[1, 3]	[1, 3]	[4, 5]
4	[4, 5]	[1, 3]	[6, 9]	[4, 5]	[1, 3]	[5, 6]	[4, 5]	[3, 4]	[4, 5]	[5, 6]	[1, 3]	[4, 5]
5	[3, 4]	[4, 5]	[6, 9]	[4, 5]	[1, 3]	[5, 6]	[4, 5]	[4, 5]	[1, 3]	[5, 6]	[1, 3]	[1, 3]
6	[3, 4]	[3, 4]	[6, 9]	[5, 6]	[3, 4]	[4, 5]	[3, 4]	[1, 3]	[1, 3]	[4, 5]	[1, 3]	[1, 3]
7	[3, 4]	[4, 5]	[6, 9]	[4, 5]	[1, 3]	[1, 3]	[4, 5]	[4, 5]	[4, 5]	[1, 3]	[1, 3]	[4, 5]
8	[4, 5]	[3, 4]	[6, 9]	[6, 9]	[3, 4]	[5, 6]	[5, 6]	[1, 3]	[1, 3]	[3, 4]	[1, 3]	[3, 4]
9	[3, 4]	[1, 3]	[3, 4]	[5, 6]	[1, 3]	[1, 3]	[5, 6]	[4, 5]	[4, 5]	[3, 4]	[3, 4]	[5, 6]
10	[4, 5]	MG	[3, 4]	[6, 9]	[1, 3]	[3, 4]	[6, 9]	[4, 5]	[3, 4]	[1, 3]	[3, 4]	[5, 6]

Table C.15: DM2: Linguistic variables of supplier assessment for problem 1

Supplier	Sub-criteria											
	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12
1	F	MP	G	MG	MP	MG	F	F	F	MP	MP	F
2	MP	F	G	F	P	MG	F	F	F	P	P	MP
3	P	P	MP	F	P	MG	MP	MP	P	MP	P	MP
4	F	MP	MP	F	MP	MP	MG	P	MG	MG	P	F
5	F	P	P	G	P	F	G	F	P	F	MP	MG
6	P	MG	F	MG	MP	MG	MP	F	P	MP	P	MP
7	P	MG	G	MG	P	P	G	F	P	MP	P	MG
8	MP	MP	P	MG	P	MG	MP	P	MG	P	MP	MP
9	P	F	F	G	MP	F	F	P	MG	MG	MP	MG
10	MP	MP	MP	F	MP	F	MG	MP	MP	P	P	MP

Table C.16: DM2: Interval values of supplier assessment for problem 1

Supplier	Sub-criteria											
	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12
1	[4, 5]	[3, 4]	[6, 9]	[5, 6]	[3, 4]	[5, 6]	[4, 5]	[4, 5]	[4, 5]	[3, 4]	[3, 4]	[4, 5]
2	[3, 4]	[4, 5]	[6, 9]	[4, 5]	[1, 3]	[5, 6]	[4, 5]	[4, 5]	[4, 5]	[1, 3]	[1, 3]	[3, 4]
3	[1, 3]	[1, 3]	[3, 4]	[4, 5]	[1, 3]	[5, 6]	[3, 4]	[3, 4]	[1, 3]	[3, 4]	[1, 3]	[3, 4]
4	[4, 5]	[3, 4]	[3, 4]	[4, 5]	[3, 4]	[3, 4]	[5, 6]	[1, 3]	[5, 6]	[5, 6]	[1, 3]	[4, 5]
5	[4, 5]	[1, 3]	[1, 3]	[6, 9]	[1, 3]	[4, 5]	[6, 9]	[4, 5]	[1, 3]	[4, 5]	[3, 4]	[5, 6]
6	[1, 3]	[5, 6]	[4, 5]	[5, 6]	[3, 4]	[5, 6]	[3, 4]	[4, 5]	[1, 3]	[3, 4]	[1, 3]	[3, 4]
7	[1, 3]	[5, 6]	[6, 9]	[5, 6]	[1, 3]	[1, 3]	[6, 9]	[4, 5]	[1, 3]	[3, 4]	[1, 3]	[5, 6]
8	[3, 4]	[3, 4]	[1, 3]	[5, 6]	[1, 3]	[5, 6]	[3, 4]	[1, 3]	[5, 6]	[1, 3]	[3, 4]	[3, 4]
9	[1, 3]	[4, 5]	[4, 5]	[6, 9]	[3, 4]	[4, 5]	[4, 5]	[1, 3]	[5, 6]	[5, 6]	[3, 4]	[5, 6]
10	[3, 4]	[3, 4]	[3, 4]	[4, 5]	[3, 4]	[4, 5]	[5, 6]	[3, 4]	[3, 4]	[1, 3]	[1, 3]	[3, 4]

Table C.17: DM1: Linguistic variables of supplier assessment for problem 2

Supplier	Sub-criteria											
	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12
1	P	F	MG	G	P	P	G	F	P	MP	MP	P
2	MP	P	MP	MP	MP	P	MP	MP	MP	F	MP	F
3	MP	MP	F	MG	MP	MG	G	MP	MP	MP	P	P
4	F	MP	F	MP	MP	MG	F	MP	F	MP	MP	MG
5	MP	MP	MG	G	P	P	MG	F	F	F	MP	MG
6	MP	MG	MG	MP	MP	MP	MG	F	MG	P	MP	P
7	F	P	MG	F	MP	P	MG	P	MP	P	MP	P
8	F	P	MG	G	MP	MG	MP	F	P	MP	MP	P
9	F	MG	MG	MG	MP	P	F	F	F	P	MP	P
10	P	F	F	MG	MP	P	MG	P	P	MP	P	MP

Table C.18: DM1: Interval values of supplier assessment for problem 2

Supplier	Sub-criteria											
	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12
1	[1, 3]	[4, 5]	[5, 6]	[6, 9]	[1, 3]	[1, 3]	[6, 9]	[4, 5]	[1, 3]	[3, 4]	[3, 4]	[1, 3]
2	[3, 4]	[1, 3]	[3, 4]	[3, 4]	[3, 4]	[1, 3]	[3, 4]	[3, 4]	[3, 4]	[4, 5]	[3, 4]	[4, 5]
3	[3, 4]	[3, 4]	[4, 5]	[5, 6]	[3, 4]	[5, 6]	[6, 9]	[3, 4]	[3, 4]	[3, 4]	[1, 3]	[1, 3]
4	[4, 5]	[3, 4]	[4, 5]	[3, 4]	[3, 4]	[5, 6]	[4, 5]	[3, 4]	[4, 5]	[3, 4]	[3, 4]	[5, 6]
5	[3, 4]	[3, 4]	[5, 6]	[6, 9]	[1, 3]	[1, 3]	[5, 6]	[4, 5]	[4, 5]	[4, 5]	[3, 4]	[5, 6]
6	[3, 4]	[5, 6]	[5, 6]	[3, 4]	[3, 4]	[3, 4]	[5, 6]	[4, 5]	[5, 6]	[1, 3]	[3, 4]	[1, 3]
7	[4, 5]	[1, 3]	[5, 6]	[4, 5]	[3, 4]	[1, 3]	[5, 6]	[1, 3]	[3, 4]	[1, 3]	[3, 4]	[1, 3]
8	[4, 5]	[1, 3]	[5, 6]	[6, 9]	[3, 4]	[5, 6]	[3, 4]	[4, 5]	[1, 3]	[3, 4]	[3, 4]	[1, 3]
9	[4, 5]	[5, 6]	[5, 6]	[5, 6]	[3, 4]	[1, 3]	[4, 5]	[4, 5]	[4, 5]	[1, 3]	[3, 4]	[1, 3]
10	[1, 3]	[4, 5]	[4, 5]	[5, 6]	[3, 4]	[1, 3]	[5, 6]	[1, 3]	[1, 3]	[3, 4]	[1, 3]	[3, 4]

Table C.19: DM2: Linguistic variables of supplier assessment for problem 2

Supplier	Sub-criteria											
	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12
1	F	F	G	MP	MP	MP	G	F	P	MP	P	MG
2	MP	P	MP	G	MP	F	G	P	MP	MG	P	F
3	MP	MP	F	MG	P	MG	G	MP	MG	P	P	MG
4	MP	MG	F	MG	MP	MG	MP	F	MG	MG	P	F
5	F	F	P	MG	MP	P	MP	F	F	P	MP	MG
6	MP	P	G	G	P	MP	P	F	MG	P	P	F
7	F	MG	MG	MG	P	P	G	F	MP	P	MP	MP
8	F	F	F	MG	MP	MG	P	P	F	MG	MP	F
9	F	P	MP	G	MP	P	F	MP	F	MP	P	MG
10	F	MG	F	F	MP	P	F	F	F	F	MP	P

Table C.20: DM2: Interval values of supplier assessment for problem 2

Supplier	Sub-criteria											
	SC1	SC2	SC3	SC4	SC5	SC6	SC7	SC8	SC9	SC10	SC11	SC12
1	[4, 5]	[4, 5]	[6, 9]	[3, 4]	[3, 4]	[3, 4]	[6, 9]	[4, 5]	[1, 3]	[3, 4]	[1, 3]	[5, 6]
2	[3, 4]	[1, 3]	[3, 4]	[6, 9]	[3, 4]	[4, 5]	[6, 9]	[1, 3]	[3, 4]	[5, 6]	[1, 3]	[4, 5]
3	[3, 4]	[3, 4]	[4, 5]	[5, 6]	[1, 3]	[5, 6]	[6, 9]	[3, 4]	[5, 6]	[1, 3]	[1, 3]	[5, 6]
4	[3, 4]	[5, 6]	[4, 5]	[5, 6]	[3, 4]	[5, 6]	[3, 4]	[4, 5]	[5, 6]	[5, 6]	[1, 3]	[4, 5]
5	[4, 5]	[4, 5]	[1, 3]	[5, 6]	[3, 4]	[1, 3]	[3, 4]	[4, 5]	[4, 5]	[1, 3]	[3, 4]	[5, 6]
6	[3, 4]	[1, 3]	[6, 9]	[6, 9]	[1, 3]	[3, 4]	[1, 3]	[4, 5]	[5, 6]	[1, 3]	[1, 3]	[4, 5]
7	[4, 5]	[5, 6]	[5, 6]	[5, 6]	[1, 3]	[1, 3]	[6, 9]	[4, 5]	[3, 4]	[1, 3]	[3, 4]	[3, 4]
8	[4, 5]	[4, 5]	[4, 5]	[5, 6]	[3, 4]	[5, 6]	[1, 3]	[1, 3]	[4, 5]	[5, 6]	[3, 4]	[4, 5]
9	[4, 5]	[1, 3]	[3, 4]	[6, 9]	[3, 4]	[1, 3]	[4, 5]	[3, 4]	[4, 5]	[3, 4]	[1, 3]	[5, 6]
10	[4, 5]	[5, 6]	[4, 5]	[4, 5]	[3, 4]	[1, 3]	[4, 5]	[4, 5]	[4, 5]	[4, 5]	[3, 4]	[1, 3]

Appendix D. Criteria Weight

Table D.21: Weight of criteria and sub-criteria

Criteria	Criteria Weight	Sub-criteria	Sub-Criteria Weight	Sub-Criteria Global Weight (w)	Priority
Service (C1)	0.567	Technical support (SC1)	0.273	0.155	2
		Information sharing (SC2)	0.067	0.038	8
		Warranty & claim policy (SC3)	0.503	0.285	1
		Capabilities (SC4)	0.156	0.089	5
Relationship (C2)	0.077	Honesty (SC5)	0.593	0.046	6
		Reputation (SC6)	0.242	0.019	9
		Trust & partnership (SC7)	0.134	0.010	11
		Ease of communication (SC8)	0.030	0.002	12
Flexibility (C3)	0.356	Product mix flexibility (SC9)	0.125	0.045	7
		Volume flexibility (SC10)	0.407	0.145	4
		Process flexibility (SC11)	0.041	0.015	10
		Service flexibility (SC12)	0.427	0.152	3

Appendix E. Input Parameters

Table E.22: Input parameters for slow movers

Parameters	Values	Units
Plant, $i \in I$		
Demand	λ_i : U(40, 100)	unit/year
Setup costs	o_i : U(500, 1000)	\$/order
Holding costs	h_i : U(10, 15)	\$/unit/year
Shortage costs	s_i : U(30, 50)	\$/unit/year
Imperfect items' holding costs	h'_i : U(30, 50)	\$/unit/year
External failure costs	a_i : U(5, 7.5)	\$/unit
Location	: [U(0, 500), U(0, 500)]	
Supplier, $j \in J$		
Supply capacity	b_j : U(100, 300)	unit
Imperfect rate	k_j : U(0.10, 0.20)	
Vehicle capacity	u_j : U(60, 90)	unit/vehicle
Disruption frequency	θ_j : U(1, 7)	days
Disruption length	v_j : U(0.5, 2)	days
Contractual costs	f_j : U(10000, 17000)	\$
Unit purchasing costs	c_j : U(25, 60)	\$/unit
Location	: [U(0, 500), U(0, 500)]	
Plant-Supplier, $i \in I, j \in J$		
Fixed transportation costs	p_{ij} : U(250, 500)	\$/order/vehicle
Variable transportation costs	r_{ij} : U(0.75, 3)	\$/mile/vehicle
Lead time	l_{ij} : $\left(\frac{U(1,2)}{60}\right)d_{ij}$	hours

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